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United States Patent Application Publication

20250283574

Kind Code

A1

Publication Date

September 11, 2025

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TRIPOD WITH A SEALED LEG LOCK

Abstract

A locking mechanism for a tripod.

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Appl. No.: 19/219276

Filed: May 27, 2025

Related U.S. Application Data

parent US continuation 18625993 20240403 parent-grant-document US 12338946 child US 19219276

parent US continuation 18320808 20230519 parent-grant-document US 11976779 child US 18625993

parent US continuation 17352031 20210618 parent-grant-document US 11698163 child US 18320808

parent US continuation 16595082 20191007 parent-grant-document US 11067216 child US 17352031

us-provisional-application US 62788256 20190104

us-provisional-application US 62826623 20190329

us-provisional-application US 62864365 20190620

Publication Classification

Int. Cl.: F16M11/32 (20060101)

U.S. Cl.:

CPC F16M11/32 (20130101); F16M2200/027 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/625,993, filed Apr. 3, 2024, which is a continuation of U.S. patent application Ser. No. 18/320,808, filed May 19, 2023, now U.S. Pat. No. 11,976,779, issued May 7, 2024, which is a continuation of U.S. patent application Ser. No. 17/352,031, filed Jun. 18, 2021, now U.S. Pat. No. 11,698,163, issued Jul. 11, 2023, which is a continuation of U.S. patent application Ser. No. 16/595,082, filed Oct. 7, 2019, now U.S. Pat. No. 11,067,216, issued Jul. 20, 2021, which claims the benefit of U.S. Provisional Patent Application No. 62/864,365, filed Jun. 20, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/826,623, filed Mar. 29, 2019, which claims the benefit of U.S. Provisional Patent Application No. 62/788,256, filed Jan. 4, 2019, which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to a portable support apparatus for equipment, such as a tripod, with a sealed leg lock. For convenience, the term “tripod” is used as a generic identifier for a more broad class of equipment supports, which may have any number of legs, e.g., a support with one, two, three, four or more legs.

[0003] Tripods have been used for many years. Typically, tripods include an elevated mounting surface, upon which equipment may be secured, such as a camera, telescope, musical equipment, surveying equipment, antenna, etc. The elevated mounting surface is supported by a plurality of legs, which in the case of a tripod is typically three. Other numbers of legs may likewise be used, such as a single leg monopod.

[0004] Tripods are designed to provide some optimal balance between a set of features. For example, one feature of a tripod is stability, as it is often desirable to insure that shaking or other movement of equipment supported by the tripod is minimized so that, for example, blur does not occur in an image taken by a camera supported on the tripod. Tripods are often used portably, hence, it is often desired to minimize the weight and/or dimensions of the tripod so that it may be carried over great distances. Also, it is often desirable to include a locking mechanism between different segments of the legs that are efficient to operate while being suitable for use in hostile environments, such as sandy environments and ocean environments.

[0005] Many existing tripods are vertically collapsible with radially extendible legs. For example, a tripod may have three leg assemblies, with each leg assembly pivotally connected to the elevated equipment support so that each leg may be pivoted outwards when in use, and inwards when not in use. Also, each leg assembly may have multiple elements that selectively slide axially relative to each other. For example, a multi-stage tripod may have first, second, and third segments for each leg. To adjust the height of the tripod, the first and second leg segments are slidably movable relative to each other and the second and third leg segments are also slidably movable relative to each other. Each pair of legs may be selectively locked and unlocked at different positions. Tripods also may include an adjustable central column to change the height at which equipment is supported by the tripod at a fixed extension of the tripod legs. Further, the tripod legs typically include locking detents so that each of the tripod legs may be positioned at one of a plurality of fixed angular orientations. The telescopic tube structure may comprise an outer tube, an inner tube

elastically sleeved by the outer tube, and a locking structure for locking two adjacent connection tubes into an integral piece.

[0006] The telescopic tubes are usually used on tripods of photographic equipment. A foot tube of a tripod is usually in the form of a telescopic tube, in order to provide a watertight locking structure of the foot tube. Chinese patent CN201651699U discloses a locking waterproof structure of leg tubes, comprising a threaded sleeve tube, a rubber sleeve, a locking part, and a locking knob component. The upper portion of the threaded sleeve tube is connected with an upper tube of a tripod leg tube. A holding cavity with the opened lower end is formed between the lower portion of the threaded sleeve tube and a lower tube of the tripod leg tube. The locking part is sleeved onto the lower tube and arranged in the holding cavity. The rubber sleeve is sleeved on the upper portion of the threaded sleeve tube. The locking knob component comprises a lock ring sleeve and a rubber part coated outside the lock ring sleeve. The lock ring sleeve is fittingly connected with the threaded sleeve tube in a thread manner and the bottom end of the rubber part is provided with a rubber part converging portion which is in interference fit with the lower tube. The thickness of the rubber sleeve meets the requirement that the upper end of the inner wall of the lock ring sleeve and the outer wall of the rubber sleeve are in interference fit after the rubber sleeve is locked through rotating the locking knob component. The waterproof of the lower end of the locking waterproof structure of leg tubes is achieved by providing a rubber sleeve at the lower end face of the lock ring sleeve, the sealing between the locking ring sleeve and the lower tube is achieved by the interference fit between the rubber part converging portion and the lower tube, during the telescopic adjustment between the upper tube and the lower tube. The rubber part converging portion is easy to be abraded, therefore the interference fit is easy to be destroyed, which eventually affects the sealing between the rubber part and the lower tube, thus, leads to a poor waterproof effect, besides, the exposed rubber part may lead to an external abrasion and corrosion, even can be peeled off from the end face, so that the waterproof performance of the locking structure is affected and even lost.

[0007] Unfortunately, in dirty and/or salty water conditions fouling tends to decrease the lockup available with the locking mechanism.

[0008] The foregoing and other objectives, features, and advantages of the invention will be more readily understood upon consideration of the following detailed description of the invention, taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0009] FIG. 1 illustrates a tripod.

[0010] FIG. 2 illustrates a pair of leg segments in an extended position.

[0011] FIG. 3 illustrates a pair of leg segments in a retracted position.

[0012] FIG. 4 illustrates a partial view of a leg segment with a leg yoke.

[0013] FIG. 5 illustrates a leg yoke and pair of sleeves.

[0014] FIG. 6 illustrates a cross sectional view of the upper portion of the leg segment and leg yoke.

[0015] FIG. 7 illustrates a foot for a leg segment.

[0016] FIG. 8 illustrates the foot of FIG. 7 together with the leg segment.

[0017] FIG. 9 illustrates a cross sectional view of the foot and leg segment of FIG. 8.

[0018] FIG. 10 illustrates a cross sectional view of the leg segment, the foot, a locking member, and the leg yoke.

[0019] FIG. 11 illustrates a cross sectional view of the locking member.

[0020] FIG. 12 illustrates a pictorial view of a knob tread of the locking member.

[0021] FIG. **13** illustrates a pictorial view of a knob substrate of the locking member.
[0022] FIG. **14** illustrates another pictorial view of the knob substrate of the locking member.
[0023] FIG. **15** illustrates a pictorial view of a collet ramp of the locking member.
[0024] FIG. **16** illustrates another pictorial view of the collet ramp of the locking member.
[0025] FIG. **17** illustrates a pictorial view of a collet lock of the locking member.
[0026] FIG. **18** illustrates a pictorial view of a collet lock of a pair of leg segments.
[0027] FIG. **19** illustrates a pictorial view of a collet lock of the locking member.
[0028] FIG. **20** illustrates a pictorial view of a circular ring of the locking member.
[0029] FIG. **21** illustrates a pictorial view of the collet lock and the circular ring of the locking member.

[0030] FIG. **22** illustrates a cross sectional view of another locking member.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENT

[0031] FIG. **1** shows an exemplary generic tripod **10**. Though features are discussed in reference to the depicted tripod, it should be understood that the features may be implemented in an equipment support having any number of legs (or leg). The tripod **10** may include a platform **12** for detachably securing equipment to the tripod **10** using a set screw or other attachment mechanism. The tripod **10** may include a plurality of legs **14** to support the platform **12** at an elevated position. The platform **12** may be detachably mounted within an apex **18** of the tripod, which may be a ring into which the platform **12** may be secured. Each of the legs **14** may include a plurality of telescoping leg sections **14A**, **14B**, **14C** and **14D**, that nest together in a retracted position by releasing an appropriate locking mechanism **16**. Although the platform **12** is shown as having a fixed vertical position with respect to the apex **18**, it is understood that the platform **12** may be supported by a vertically-extensible column movably (e.g. slidably) engaged with the apex **18**. The platform **12** may include a spirit level. Each of the legs **14** preferably pivots about a respective axis proximate the apex **18**. Through a selective choice of respective angles between each of the legs **14** and the apex **18**, along with a desired telescoped length of each respective leg, equipment attached to the tripod **10** may be supported on a wide variety of terrain.

[0032] FIG. **2** illustrates a tripod leg with two leg sections **20A**, **20B** that nest together with one another. Each of the two leg sections **20A**, **20B** are preferably centrally axially aligned with one another. The end of leg section **20A** may include a foot **22**, having any suitable structure, such as a crenated foot, a spike, or a rounded end. The two leg sections **20A**, **20B** are adjustably slidably secured together by the locking mechanism **24**. By rotation of the locking mechanism **24** in a first direction, the leg sections **20A**, **20B** may be readily slid with respect to one another. By rotation of the locking mechanism **24** in a second direction, opposite the first direction, the leg sections **20A**, **20B** are inhibited from being readily slid with respect to one another. The end of leg section **20B** may include a leg yoke **26** that is movably attached to the apex **18**.

[0033] FIG. **2** illustrates the two leg sections **20A** and **20B** in a generally extended position, and may be inhibited from being readily slid with respect to one another using the locking mechanism **24**. FIG. **3** illustrates the two leg sections **20A** and **20B** in a generally retracted position, and may be inhibited from being readily slid with respect to one another using the locking mechanism **24**. Intermediate positions of the two leg sections **20A** and **20B** may likewise be selected and secured, as desired, using the locking mechanism **24**.

[0034] Referring to FIG. **4**, a partial view of FIG. **3**, illustrates the two leg sections **20A** and **20B**, together with the leg yoke **26** with the locking mechanism **24** not shown. A pull tab **29** is illustrated, together with an appropriate spring mechanism (not shown), which is used for a ratcheting mechanism for securing the leg at different angular orientations with respect to the apex **16**.

[0035] Referring also to FIG. **5**, the leg yoke **26** is illustrated together with a pair of anti-twist sleeves **30A** and **30B**. The pair of anti-twist sleeves **30A** and **30B** fit within the interior of the tube a substantial distance, being in face-to-face engagement with the interior surface of the tube, where

the tube is bonded to the leg yoke **26**. The pair of anti-twist sleeves **30A** and **30B** may also extend a substantial distance **32A** and **32B** external to a lower portion **34** of the leg yoke **26**.

[0036] Referring also to FIG. **6**, the pair of anti-twist sleeves **30A** and **30B** are in face-to-face engagement with the interior of the leg section **20B**. The interior surface of the leg yoke **26** may include a pair of spaced apart recessed portions **38A** and **38B**, which act as adhesive injection spillover chambers, that preferably extend around the entire interior periphery in a ring-like fashion, to secure the leg section **20B** to the leg yoke **26**. The spacing between the anti-twist sleeves, which are preferably slightly flexible, assist in removal and insertion of the anti-twist sleeves within the leg yoke **26** and leg section **20B**.

[0037] When the two leg sections **20A** and **20B** are extended or retracted the amount of space enclosed therein tends to increase or decrease, respectively. With the two leg sections **20A** and **20B** substantially sealed by the foot **22**, the locking mechanism **24**, and the yoke **26**, it is desirable to include an opening defined therein to permit a free-flow of air to adjust the pressure of the air enclosed by the two leg sections **20A** and **20B**. Referring to FIG. **5** and to FIG. **6**, a slit **50** may be defined by the yoke **26** that extends from the interior of the yoke **26**, which is in fluid communication with the interior of the leg sections **20A** and **20B**, to the exterior of the yoke **26**, which is in fluid communication with the external environment. The slit **50** may be any suitable type of opening and/or passage defined by the yoke **26**, or the yoke **26** in combination with other structures. With the slit **50** being located just below the apex, and hence high away from likely sources of fouling, such as dirt and salt water, reduces the likelihood of damage to the tripod from such fouling. The slit **50** permits air exchange between the interior and the exterior of the two leg sections **20A** and **20B**, permitting the leg sections to more freely move without significant backpressure. In addition, while the slit **50** may be forwardly facing, it is preferably rearwardly facing so that it directs air away from a ratcheting pull tab mechanism, which tends to include relatively high tolerance components that may be rendered inoperable if occasionally sprayed with wet, salty air, from inside the two leg sections **20A** and **20B**.

[0038] Referring to FIG. **7**, the foot **22** may include a bowl foot **60**. The bowl foot **60** preferably has a bowl shaped exterior surface, but may include any other type of shape, such as a spike or a ring. The foot **22** may include a tee nut with **4** prongs **62**, which may be detachably engaged with the bowl foot **60**. The tee nut **62** may be detachably engaged with a tripod foot receiver **64** using a set screw **66**, which may be threadably engaged with threads in the foot **22** using the set screw **66**, passing through the tee nut **62**. Referring also to FIG. **8**, the leg segment **20A** may be engaged with a tripod foot receiver **64** with the lower portion thereof of the leg segment **20A** in face-to-face engagement with the interior surface of the tripod foot receiver **64**, as shown in FIG. **9**. It is noted that the foot together with the leg segment forms a structure that is completely sealed which prohibits fluids to enter or to exit the lower portion of the tripod leg.

[0039] Referring to FIG. **10**, a cross sectional view of a leg assembly is shown with the foot **22**, the two leg segments **20A** and **20B** in a generally extended position, the locking mechanism **24**, and the yoke **26**. Referring to FIG. **11**, an enlarged view of a portion **70** of the locking mechanism **24** is illustrated. The locking mechanism **24** encircles the two leg sections **20A** and **20B** and by rotation of a knob tread **100** (see FIG. **11**) in a first direction, the two leg sections **20A** and **20B** are inhibited from sliding with respect to one another, and by rotation of the knob tread **100** (see FIG. **11**) in a second direction opposite the first direction, the two leg sections **20A** and **20B** are released from being inhibited from sliding with respect to one another. Referring to FIG. **12**, the knob tread **100** may include a set of raised ribs **102** for a better grip by the user. The interior surface **104** of the knob tread **100** is preferably molded over a heavily knurled surface of an exterior surface **112**. Preferably, the knob tread **100** is a flexible rubberized material.

[0040] Referring also to FIG. **13** and FIG. **14**, the locking mechanism **24** includes a knob substrate **110**. The knob substrate **110** includes the exterior surface **112** thereof that defines a pair of grooves **114** and **116**. The shoulders **114** and **116** serve as shutoff seals for the tread over-molding process

and also as axial external retainers for the tread **100** once molded. Also, the knob tread **100** includes a solid shoulder that is fully recessed within the pair of shoulders **114** and **116**. The knob substrate **110** may include an internal surface **118** that defines a set of threads **120**. The threads are engaged with threads **132** defined by a collet ramp **130**. The knob substrate **110** may define a back circular recess **122** that is sized to receive a back O-ring **150** that is slightly larger in depth and/or width than the back circular recess **122**.

[0041] Referring to FIG. **15**, the collet ramp **130** may define the exterior set of threads **132** which are rotatably engaged with the threads **120** of the knob substrate **110**. The exterior surface **134** of the collet ramp **130** defines a forward circular recess **136** that is sized to receive a forward O-ring **154** that is slightly larger in depth and/or width than the forward circular recess **136**. The forward portion **140** of the collet ramp **130** preferably has a higher elevation than the rearward portion **142** of the collet ramp **130**. The surface of the forward portion **140** serves both to allow for an O-ring groove and to add structural strength in the tripod leg section sub-assembly. The surface of the rearward portion **142** is preferably near co-radial with the forward circular recess **136**, such that they form an inner half of seals for a pair of O-rings **150**, **154**, where the forward seal gets compressed from the outside diameter and the rearward seal gets compressed from the inner diameter. Also, preferably the surface of the rearward portion **142** is at substantially the same elevation as the surface of the forward circular recess **136**. The collet ramp **130** may be secured to the exterior surface of the leg segment **20B**, such as using an adhesive, so that the collet ramp **130** is maintained in a fixed relationship with respect to the leg segment **20B**.

[0042] The forward O-ring **154**, which is deformable, forms a substantially waterproof seal between the collet ramp **130** and the knob substrate **110** thereby keeping water and other debris out of the locking mechanism **24** from above and thereby the leg assembly. With a relatively low compression and a relatively slow reciprocating motion permits a relatively soft durometer to be used for the forward O-ring **154**, and thereby results in minimal perceivable drag on the twist action of the locking mechanism **24**.

[0043] The back O-ring **150**, which is deformable, forms a substantially waterproof seal between the collet ramp **130** and the knob substrate **110** thereby keeping water and other debris out of the locking mechanism **24** from below and internally, and thereby the leg assembly. With a relatively low compression and a relatively slow reciprocating motion permits a relatively soft durometer to be used for the back O-ring **150**, and thereby results in minimal perceivable drag on the twist action of the locking mechanism **24**.

[0044] With the pair of O-rings **150** and **154**, one toward the forward portion of the locking mechanism **24** and one toward the back portion of the locking mechanism **24**, the locking mechanism **24** forms a substantially waterproof seal between the collet ramp **130** and the knob substrate **110**, even as the knob substrate **110** moves with respect to the collet ramp **130** and even as the leg segments **20A** and **20B** move with respect to one another.

[0045] The combination of the collet ramp **130** and the knob substrate **110** preferably define an open region **144** so that upon rotation of the knob substrate **110** with respect to the collet ramp **130**, the knob substrate **110** slides based upon the threads in a direction in line with the leg segments **20A** and **20B** with respect to the collet ramp **130**. While the knob substrate **110** slides with respect to the collet ramp **130** over its range of motion, the forward O-ring **154** and the back O-ring **150** both maintain the substantially waterproof seal.

[0046] Referring also to FIG. **16**, the collet ramp **130** includes a sloped surface **160** defined by the interior of the collet ramp **130**. Referring to FIG. **17**, a collet lock **180** may include a set of tapered fingers **182** that define a sloped surface **184**. The slope of the sloped surface **184** of the tapered fingers **182** preferably substantially matches the slope of the sloped surface **160** of the collet ramp **130**. Preferably, the sloped surface is between 5-10 degrees, and more preferably between 6-7 degrees, to provide for fast lock and unlock speeds, in a manner not likely to cause the locking member **24** to become stuck in the process. In this manner, the sloped surfaces **160** and **184** may

slide with respect to one another with a face-to-face engagement. The collet lock **180** preferably includes a circular depression **186** that matches an exterior protrusion **188** of the knob substrate **110**.

[0047] The collet lock **180** is maintained in a fixed relationship with respect to the knob substrate **110** with the circular depression **186** and the exterior protrusion **188**. The collet ramp **130** is maintained in a fixed relationship with respect to the leg segment **20B**. With the rotation of the knob tread **100**, the knob substrate **110** rotates with respect to the collet ramp **130** and are moved axially based upon the threads in line with the leg segments **20A** and **20B** which selectively increases and decreases the pressure inhibiting movement of the leg segments **20A** and **20B** with respect to one another.

[0048] Referring to FIG. **18**, a cross sectional view of a leg assembly is shown with the foot **22**, the two leg segments **20A** and **20B** in a generally retracted position, the locking mechanism **24**, and the yoke **26**.

[0049] Referring to FIG. **19**, the collet lock **180** is illustrated, with a different view than the collet lock **180** illustrated in FIG. **17**. To provide the capability of selectively keeping fluid and other particles out of the locking mechanism **24**, it is desirable to include a blocking member, detachably engaged with the collet lock **180**. Referring to FIG. **20**, one type of blocking member is a circular ring **200** that snaps into the substrate **110** into the circular recess **122**. Referring to FIG. **21**, the circular ring **200** has an inside diameter that is slightly greater than the corresponding diameter of the leg segment **20A**, so that it does not substantially rub on the exterior surface of the leg segment **20A** providing for fast telescoping of the legs. In this manner, flowing fluid will tend to be mostly diverted away, and all but the smallest silty particles will be totally blocked. Referring to FIG. **22**, a cross sectional view of a locking mechanism **24** together with the circular ring **200**, is illustrated. The circular ring **200** is readily replaceable by the user.

[0050] Another type of blocking member is a circular ring **200** that snaps into the collet lock **180**. The alternative circular ring **200** has an inside diameter that is substantially the same as the corresponding diameter of the leg segment **20A**, so that it substantially rubs on the exterior surface of the leg segment **20A** providing for substantially slower telescoping of the legs due to the increased resistance. In this manner, flowing fluid will tend to be almost all diverted away, and substantially all including substantially all of the smallest silty particles, will be totally blocked. The circular ring **200** is readily replaceable by the user.

[0051] Another type of blocking member is a v-shaped ring **210** (see FIG. **11**) that snaps into the collet lock **180**. The alternative v-shaped ring **210** has an inside diameter that is substantially the same as the corresponding diameter of the leg segment **20A**, so that it substantially rubs on the exterior surface of the leg segment **20A** providing for substantially slower telescoping of the legs due to the increased resistance. Moreover, the v-shaped cross-sectional tends to result in an externally directed force against the exterior of the leg segment **20A**. In this manner, flowing fluid will tend to be almost all diverted away, and substantially all including substantially all of the smallest silty particles, will be totally blocked. The v-shaped ring **210** is readily replaceable by the user.

[0052] As previously described, there are two separate seal mechanisms that may be included, either of which may be omitted, if desired. Referring to FIG. **11**, in general, the seals that include O-ring **150** and O-ring **154** inhibit contaminants from reaching the threaded members **130** and **132** and also inhibit the user from actuating the lock. The O-ring **154** is typically sufficient to substantially keep contaminants out of the mechanism without the need for the additional O-ring **150** if the v-shaped ring **210** inhibited all (or substantially all) contaminants from entering through the vent and/or the foot receiver. The O-ring **150** may be included as an additional fail safe in order to inhibit the threads from becoming fouled in the event that contamination enters through the vent, the foot receiver, or if the v-shaped ring **210** does not provide a sufficient seal (as is the case when using ring **200** of FIG. **20** instead of ring **210** of FIG. **11**).

[0053] As previously described, the v-shaped ring **210** (see FIG. **11**) and/or the v-shaped ring **200** (see FIG. **20**) inhibit contaminants from reaching the collet (see, FIG. **11**, element **180**), anti-twist twist sleeves (see, FIG. **5**, elements **32A** and **32B**), and getting stuck inside the inner tubes (see, FIG. **4**, element **20A**). In practice, the threads tend to limit the intrusion of contaminants from entering the tube assembly so that the O-ring **150** and the O-ring **154** may be omitted, especially if the v-shaped ring **210** is included. Over time, some contamination may pass through the threaded region but is substantially inhibited by grease included on the threads. By including a seal mechanism, this reduces fouling of the collet which can prevent the collet from collapsing on the tube and locking. In addition, by including a seal mechanism, this reduces particles from wedging between the collet and the tube, acting like little ball bearings thereby decreasing the locking friction. Further, by including a seal mechanism, this reduces fouling in other parts of the tube which may lead to trapped contaminants that leave foul odors or harden over time which inhibit full functionality.

[0054] The terms and expressions which have been employed in the foregoing specification are used therein as terms of description and not of limitation, and there is no intention, in the use of such terms and expressions, of excluding equivalents of the features shown and described or portions thereof, it being recognized that the scope of the invention is defined and limited only by the claims which follow.

Claims

1. A support suitable for supporting a device thereon comprising: (a) said support including a plurality of elongate hollow leg members supporting an apex, said apex suitable for supporting said device thereon; (b) each of said plurality of elongate hollow leg members including respective a first leg segment and a second leg segment that are slidably movable relative to one another and wherein each of said plurality of elongate hollow leg members includes a respective leg yoke at a respective terminal portion thereof; (c) a respective locking mechanism selectively inhibiting said slidably movable between said first and second leg segments for respective ones of said plurality of elongate hollow leg members, wherein said respective locking mechanism is configured to provide a substantially waterproof seal using a first member engaged therein to inhibit liquid from reaching the interior of said first leg segment and said second leg segment for respective ones of said plurality of elongate hollow leg members; (d) each of said locking mechanisms includes a liquid inhibiting structure with a first recess defined therein and said first member engaged therein, where said first member is resiliently deformable, where said first member extends proud of said first recess, wherein said liquid inhibiting structure is proximate a first end of said respective locking mechanism relative to an opposing end of said respective locking mechanism, wherein said first recess at least one of (i) opens up toward said at least one elongate hollow leg member and (ii) opens up away from said at least one elongate leg member, wherein locking said locking mechanism compresses said first member, wherein unlocking said locking mechanism decompresses said first member; (e) wherein each of said respective yokes defines a respective yoke opening therein that permits free-flow of air between an inside of each of respective said one of said plurality of elongate hollow leg members and an exterior of respective said yoke when respective said first leg segment and said second leg segment are slidably moved relative to one another thereby changing the volume of space defined within respective ones of said plurality of elongate hollow leg members; (f) wherein each of said plurality of elongate legs is rotatable in an arch relative to said apex such that each of said plurality of elongate legs are rotatable outwardly in a manner such that said plurality of elongate legs are only capable of supporting said apex at a position above respective terminal end portions of said plurality of elongate legs in a manner such that said device is positioned at an elevation higher than any portion of said plurality of elongate legs when said respective terminal end portions of each of respective said plurality of elongate legs

are positioned on the ground, where said respective terminal end portions are at an opposite end of each of respective said plurality of elongate legs relative to said yolk, where when said respective terminal end portions of said plurality of elongate legs are positioned on the ground and supporting said apex at rising water levels relative to the ground first reach a lower end of a respective locking mechanism prior to reaching an upper end of said respective locking mechanism.

2. The support of claim 1 wherein each of said locking mechanisms said selectively inhibits respective said slidably movable by selective rotation.
3. The support of claim 1 wherein each of said plurality of elongate hollow leg members is rotatably connected to said apex, a respective slidable tab operationally engaged with said support to selectively inhibit respective said rotatably connected between each of said plurality of hollow leg members and said apex.
4. The support of claim 1 wherein each of said yokes further comprising at least one sleeve portion that is engaged in face-to-face abutment between an exterior surface of said at least one sleeve portion and an interior surface of respective said elongate hollow leg member to inhibit rotation of respective said yoke with respect to respective said elongate hollow leg member.
5. The support of claim 4 wherein respective said exterior surface of said at least one of said sleeve portion includes a pair of spaced apart recessed portions.
6. The support of claim 1 wherein each of said elongate hollow leg members is rotatably connected to said apex, a respective slidable tab operationally engaged with said support to selectively inhibit said rotatably connected between respective said elongate hollow leg member and said apex, respective said yoke opening oriented in a direction away from respective said slidable tab.
7. The support of claim 1 wherein each of said elongate hollow leg members includes a respective foot assembly at a terminal portion thereof, respective said foot assembly includes a respective foot portion that is suitable for supporting said support on a ground, respective said foot assembly includes a respective receiver portion engaged with a respective surface of respective said elongate hollow leg member, a respective tee nut together with a respective set screw detachably engaging respective said foot portion with respective said receiver portion in such a manner that when respective said foot portion is engaged with respective said receiver portion water is inhibited from entering respective said elongate hollow leg member from a location proximate with respective said foot assembly.
8. The support of claim 1 wherein one of said locking mechanism includes a knob tread with an exterior surface that defines an exterior set of raised ribs and an interior surface that is in face-to-face engagement with a knurled surface.
9. The support of claim 1 wherein one of said locking mechanism includes a knob substrate that includes an exterior surface that defines a pair of grooves and an interior surface that defines a set of knob substrate threads.
10. The support of claim 9 further comprising said one of said locking mechanism includes a collet ramp that defines a set of collet ramp threads that are rotatably engaged with said knob substrate threads.
11. The support of claim 10 further comprising said interior surface of said knob substrate defining at least one of said first recess and a second recess.
12. The support of claim 10 further comprising said collet ramp defining the other one of said first recess and a second recess.
13. The support of claim 11 further comprising said collet ramp defining the other one of said first recess and said second recess, a second resiliently deformable member maintained within the other one of said first recess and said second recess.
14. The support of claim 10 further comprising an interior surface of a collet lock defines a collet lock sloped surface where said collet lock is positioned at a location interior to said collet ramp.
15. The support of claim 14 wherein said collet lock sloped surface is defined by a plurality of sloped fingers.

- 16.** The support of claim 14 wherein said collet lock sloped surface is substantially the same slope of said collet ramp.
- 17.** The support of claim 16 wherein said collet lock is maintained in a fixed position with respect to said knob substrate, such that rotation of said knob substrate results in matching rotation of said collet lock.
- 18.** The support of claim 17 further comprising a block member that is detachably engageable with said collet lock to inhibit fluid from reaching a junction of said first leg segment and said second leg segment.
- 19.** The support of claim 1 wherein said yolk includes a latching mechanism to selectively position one of said elongate legs in a selected orientation.
- 20.** A support suitable for supporting a device thereon comprising: (a) said support including at least one elongate hollow leg member at least partially supporting an apex, said apex suitable for supporting said device thereon; (b) said at least one elongate hollow leg member including a first leg segment and a second leg segment that are slidably movable relative to one another; (c) a locking mechanism selectively inhibiting said slidably movable between said first and second leg segments, wherein said locking mechanism is configured to provide a substantially waterproof seal to prevent liquid from reaching the junction of said first leg segment and said second leg segment; (d) wherein said at least one elongate hollow leg member includes a leg yoke at a terminal portion thereof; (e) wherein said yoke defines a yoke opening therein that permits free-flow of air between an inside of said at least one elongate hollow leg member and an exterior of said yoke when said first leg segment and said second leg segment are slidably moved relative to one another thereby changing the volume of space defined within said at least one elongate hollow leg member.
- 21.** The support of claim 20 wherein said at least one elongate hollow leg member is rotatably connected to said apex, a slidable tab operationally engaged with said support to selectively inhibit said rotatably connected between said least one elongate hollow leg and said apex, said yoke opening oriented in a direction away from said slidable tab.
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